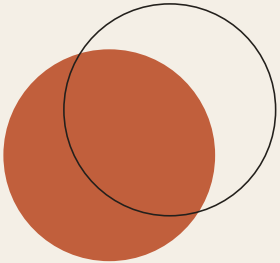
November 9, 2026

Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger



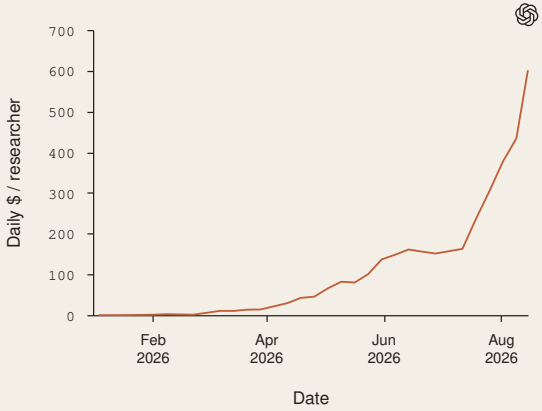


OUTLINE

I. OpenAI and Agent Usage

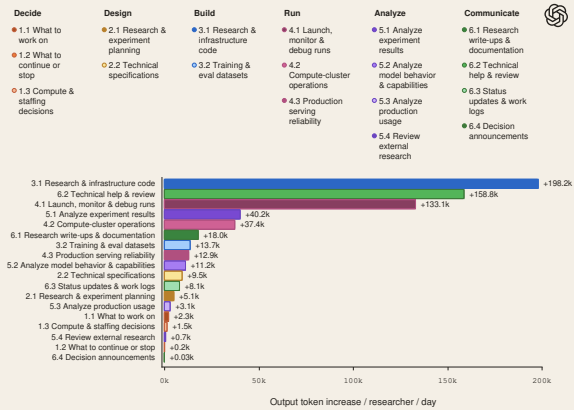
II. Owning Intelligence and Infrastructure at Lamarr

OpenAI: ∼$600 per Researcher per Day



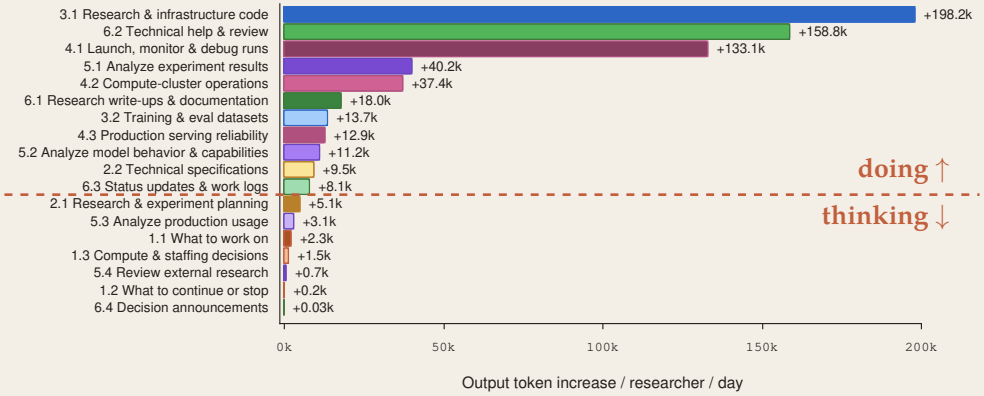
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported)

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, “Research acceleration” (self-reported)

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What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open

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Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure	Agent use is a *skill*: not everything can or should be automated (degree of automation)

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Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises`

   Avoid: Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it

II

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EXAMPLE: TRAVEL EXPENSE REPORTS

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips × 3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system), plus upkeep as rules change

	how often × time saved	time to build	
one person	$3 \times 3\,\text{h} = 9\,\text{h/year}$	200 h	**never pays**

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Lamarr, ∼300 researchers	$300 \times 9\,\text{h} = 2\,700\,\text{h/year}$	200 h	**pays >13×**

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A request to management: right now, many of us pay for these tools ourselves. Most of us stop at the 20 € plan rather than the 100 € one, so we get less out of the tools than we could. Institutional access would help us do better work and it will matter more and more for attracting and keeping good people, as other institutes, in Germany and abroad, start to provide it

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So let's talk about costs

Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
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One month of my usage, priced at API list prices

	subscription	same usage via API	factor
Claude	20 $ (Pro)	∼680 $	∼34×
Codex	20 $ (Plus)	∼145 $	∼7×

Decision 2: Self-Hosting?

In the news

"Revolut turns to self-hosted AI to cut coding costs tenfold over Claude"

Computing (UK), 2026

Decision 2: Self-Hosting? A Field Report: Pros and Cons

Hardware	Model	Quality	Speed
4 × H100	**GLM 5.3 Flash**	**Claude Sonnet**	**50–100 tokens/s**
repurposed	mainly	high effort, by feel	per user

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- Pipelines, e.g. **AutoExercises**, finding related work
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- **Private data**
- Running **agent swarms**

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What I would not use it for

- High-intelligence, low-token work: math proofs, idea finding
- Few tokens (see the OpenAI chart), so the best model stays affordable

Decision 2: Self-Hosting? A Field Report: Costs

To put the headline in perspective: energy and the Sonnet comparison are real values, sent to me by a colleague I trust.
We do not go into all the details of self-hosting.
Repurposing hardware also takes skill and neat tricks.

GLM-5.3-Flash (main model)	**per month**
energy (full load)	∼330–670 €
staff (2–3 h/week at 50 €/h)	∼430–650 €
energy + staff	**∼0.8–1.3 k€**
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Payback: 4 × H100 at ∼150 k€ pays for itself in ∼10–22 months

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**
   how often × time saved > time to build
4. **Use AI to build your own AI setup**
   Learn by doing, tailor it to your workflow, and make your daily work easier

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For the institute

1. **Automate workflows for every individual:** what never pays off for one person pays off for 300
2. **Provide your employees with agents**
3. **Consider self-hosting**

Next: David's Setup

1. We know that we **need AI agents** and what to use them for
2. David shows **his own setup**: how it looks in daily work for him, ML research specific
3. The best setup is **personal and custom**, so build your own